

Measuring the Wettability of Different Ferrofluid Concentrations Under the Influence of Magnetic Fields.

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Abstract

This experiment is aimed at studying the effects of magnetic fields on the wettability and related properties of a ferrofluid. The wettability of a fluid on a solid surface is proportional to the contact angle and height of the liquid drop and describes how the two materials maintain contact and interact. The concentration of ferrofluid, the distance between the 0.309 T neodymium magnet and droplets, and the solid surface material were the three parameters varied to capture the change in ferrofluid behaviour under different magnetic field strengths. Height, contact angle, and diameter of the drops were recorded using ImageJ software to analyse the experimental photos taken with a macro lens. The distance between the macro lens and the drop sample was kept constant to measure the height and diameter with a consistent scale. The PDMS surface produced clearer images and better data than the glass slide as the ferrofluid was less wettable at this interface. However, the results of this experiment did not show the expected trend of contact angle increasing proportionally with magnetic intensity. Furthermore, expected relations were not able to be quantified for wettability of various ferrofluid dilutions since the standard error on the data produced results that are orders of magnitude from the expected results. These results are reflected in the Chi-Squared values for a 1:0 dilution of the ferrofluid used, $\chi^2_{1:0 \text{ Angle}} = 143.98$, $\chi^2_{1:0 \text{ Height}} = 8.47$, and $\chi^2_{1:0 \text{ Diameter}} = 168.58$. These values of χ^2 are high and show that there is limited correlation between this experiments results and the expected values of angle, height, and diameter. To better measure the relationship between wettability for ferrofluid, there are several steps including automating data collection to higher precision, measuring the exact dilution of samples, and using a more standard droplet sizes.

1. Introduction

A ferrofluid is a fluid that consists of colloidal suspensions of ferromagnetic nanoparticles dispersed in a non-magnetic liquid [1] [2]. The wettability of a fluid is defined as the ability of a liquid to maintain contact with a solid interface. The level of wetting is determined by the contact angle between the surface of the solid base and the meniscus of the fluid. As illustrated in Figure 1, a small contact angle ($<90^\circ$) signifies that the fluid has high wettability, while a large contact angle ($>90^\circ$) demonstrates that the fluid has low wettability [3].

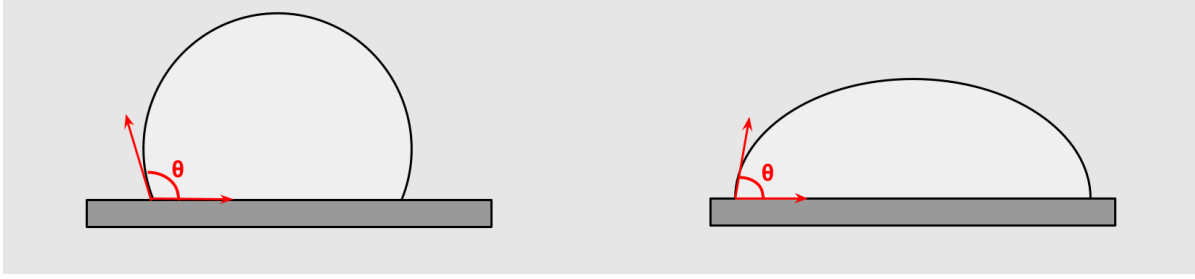


Figure 1: Examples of the contact angle and the associated wettability. The fluid on the left has a low wettability as seen by the large θ value while the fluid on the right has a high wettability as seen by its low contact angle θ [4].

The goal of this experiment was to observe and quantify how the wettability of ferrofluid changes when exposed to different strengths of non-uniform magnetic fields. The wettability of the ferrofluid can be determined using the sessile drop method. The sessile drop method involves the placement of a liquid on a smooth solid surface and measuring the angle between the solid surface and the tangent to the drop profile, in addition to the height and radius [4]. Below the ferrofluid, a neodymium magnet produces a magnetic field, altering the shape of the drop.

During this experiment, three parameters were varied: the concentration of the ferrofluid, the gradient of the magnetic field, and the slide material composition to explore the difference between wettability of ferrofluid on glass and PDMS. Previous studies had found a relationship between a rising magnetic field and the increase in drop width and a decrease in the drop height [5]. This change in drop geometry corresponds to a decrease in the wetting angle with an increasing magnetic field. From these parameters, the sessile drop method of determining the necessary measurements was done using the image processing software ImageJ. ImageJ is a commonly used graphics processing tool used mainly for measuring and annotating visual data in material scientific experiments [6]. Alternatively another option for analysis of contact angles is KRUS, a similar type processing software [7]. ImageJ produces distances in pixels, so a scale was calibrated using a known distance and determining the number of pixels per mm from this image. Since contact angle is independent of distance, no conversion was required to measure the angle of each droplet using ImageJ.

The importance of this research within the scientific community was considered when designing the experiment. Ferrofluids are used within magnetically driven devices such as setpper motors, loudspeakers and several sensors. Furthermore, the findings of the experiment related to wettability of different concentrations of ferrofluid could aid in varying heights, contact angles, and surface spread of varying diluted ferrofluid droplets could aid in the development of scientific research in microfluidics, magnetic actuation, and magnetically activated transducers [8]. A specific application of magnetically sensitive transducers relates to compact magnetic field sensors [9]. Compact magnetic sensors have a plethora of uses, but are commonly used to measure rotational velocity, linear position and angles of machines like automobiles or industrial robotics [10].

2. Model and Predictions

2.1 Surface Tension of Drops

The following equations will be used to derive the relationship between the energy balance at the corner of the sessile drop and the surface energy of the drop. Simplifications were made to make

the experimental procedure feasible within the given experimental conditions and limitations. Since this is a sessile drop, the system can be assumed to be in equilibrium therefore the surface tensions present in Figure 2 obey the relation shown in Equation 1.

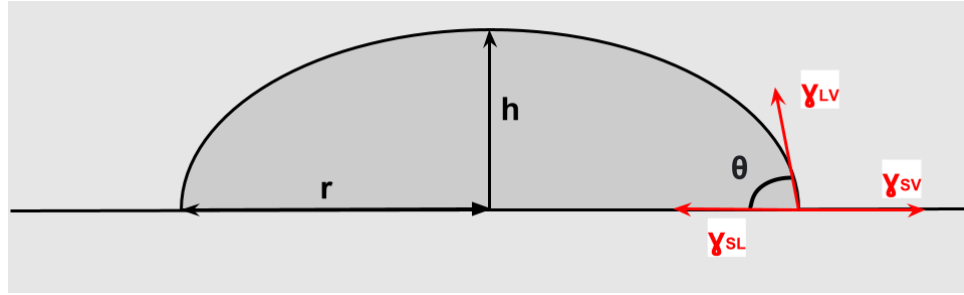


Figure 2: Equilibrium force balance at the corner of a sessile drop, related by an angle, theta.

Equation 1 describes the equilibrium tension balance of a sessile drop on a solid substrate, related by the wetting angle theta [11].

$$\gamma_{sv} = \gamma_{sl} + \gamma_{lv} \cos \theta \quad (1)$$

In Equation 1, γ_{sv} is the surface energy between the solid-vapour interface, γ_{lv} is the surface energy between the liquid-vapour interface, γ_{sl} is the surface energy between the solid-liquid interface and θ is the wetting angle associated with the drop. An energy balance can be applied to this situation where the system can be modeled as shown in Equation 2 [11].

$$\sum E_{system} = E_m + E_s \quad (2)$$

The two energy terms in the sum represent the potential energy of the fluid E_m and the change in the surface energy due to liquid wetting E_s . This can be expanded to arrive at Equation 3 [11].

$$E = \rho V g C - V(\gamma_{sv} - \gamma_{sl}) \quad (3)$$

Where V is the volume of the drop, C is the center of mass of the drop and ρ is the specific density of the given ferrofluid concentration. For the purposes of this model, it has been assumed that half of a general ellipsoid shape would accurately represent the expected dome shape of the drop. Substituting Equation 1 into Equation 3 and taking the derivative of the energy with respect to the height of the drop Equation 4 is produced [11].

$$\frac{dE}{dh} = 0 = \frac{5}{6} \rho r^2 h - \frac{2}{3} r^2 \cos(\theta) \gamma_{lv} \quad (4)$$

Rearranging the relationship for the liquid-vapour vapour surface tension of a sessile drop on a solid substrate can be achieved, as seen in Equation 5 [11].

$$\gamma_{lv} = \frac{5}{4} \frac{\rho h}{\cos(\theta)} \quad (5)$$

This relationship allows for a prediction of wettability, as the contact angle and height of the photographed drops increase, the surface energy as a measure of wettability will increase. Furthermore, as density changes with concentration, an increase in density will also increase wettability.

2.2 Intensity of Magnetic Field

The strength of the magnetic field can be calculated using Equation 6, Magnetic field strength of a rectangular magnet as a function of vertical distance above the magnet (z), derived from the Biot-Savart Law [12].

$$B(z) = \frac{B_r}{\pi} \left(\arctan \left(\frac{lw}{2z\sqrt{4z^2 + l^2 + w^2}} \right) - \arctan \left(\frac{lw}{2(t+z)\sqrt{4(t+z)^2 + l^2 + w^2}} \right) \right) \quad (6)$$

Where B_r is the unknown residual induction of the magnet, l and w are the length and width, t is the thickness, and z is the vertical distance between the magnet surface and the testing point which in this case will be the ferrofluid drop. Based on the characteristics of the magnet used in this experiment, the relationship should hold as illustrated in Figure 3.

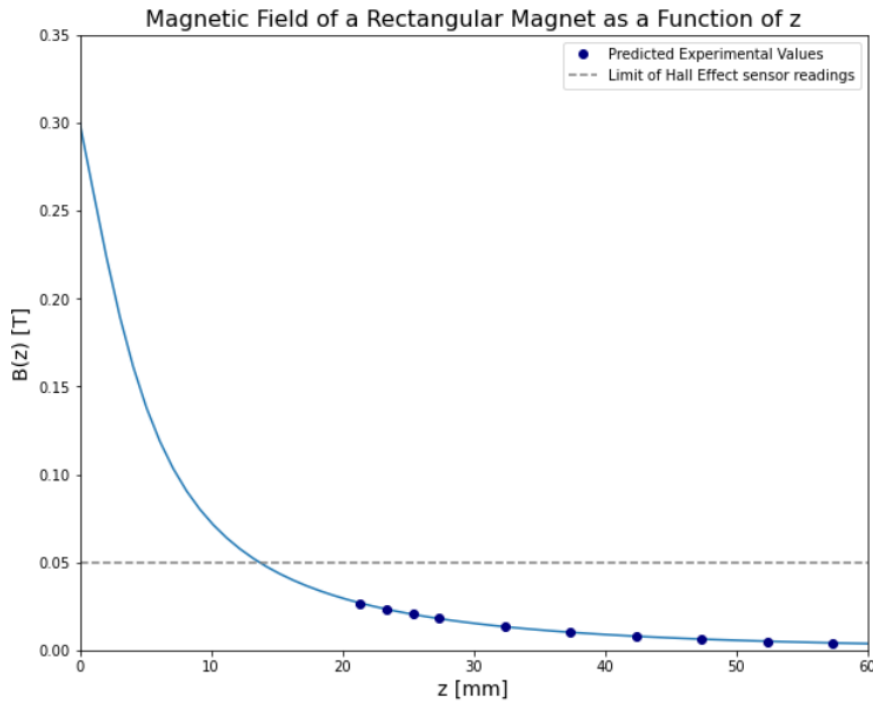


Figure 3: Magnetic field strength in Tesla modelled by Equation 6 with varying values of distance z vertically above the magnets. 10 data points will be used within the range of allowed sensor readings.

Since the magnets being used were strong neodymium magnets, they were separated using spacers, which needed to be accounted for in the modelling of this function. At a distance of zero mm from the magnets, a maximum magnetic field measurement was made using a magnetometer and found to be 0.309 ± 0.001 T. This data collection was not directly relevant to the experiment but was required to predict the theoretical curve for magnetic intensity. By substituting this known datapoint into Equation 6, the value of the residual magnetic field, B_r , was calculated to be 0.9195 ± 0.0005 T.

As the magnetic field is expected to decrease with distance z , this will change the behaviour and wetting characteristics of the ferrofluid drops. It is expected that, since magnetic force is proportional to the field strength, the contact angle and height of the drops will decrease as distance increases. The expected relationship of both contact angle and diameter of the droplets can be seen in Figure 12 which was extracted from the paper *Static Magnetowetting of Ferrofluid Drops* [5].

3. Design of Apparatus

The apparatus consisted of two stands with acrylic slabs clamped to each. Two stands for each slab were used, rather than both slabs being connected to one rod, to reduce the interference adjusting the distances would have. Six neodymium magnets were used in the apparatus, these were placed in the bottom acrylic slab, the acrylic was machined to have a rectangular indent to hold the magnets and ensure they did not move over the course of the experiment for consistent vertical magnetic intensity. The top acrylic was also machined to hold a rectangular microscope slide. The acrylic slab set up is illustrated in Figure 4. The diagram shows the top and bottom acrylic, while the shaded rectangular regions show the machined indents that hold the microscope slide (on the top) and magnets (on the bottom). The dotted line shows that they are aligned so the centers of the machined indents overlap. The top acrylic piece was carefully leveled so the ferrofluid drop would be on a flat surface and not slide in any direction during data collection. The clamps and stands were omitted from Figure 4 for simplicity.

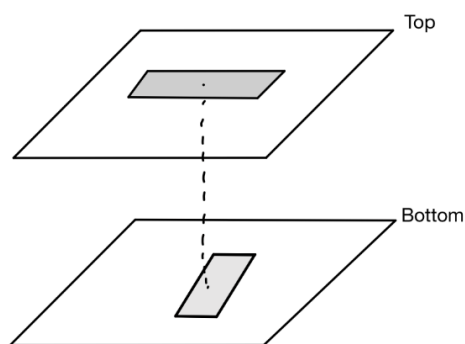


Figure 4: Diagram of how the acrylic pieces sit atop each other. The clamps and rest of the set up is omitted for simplicity. The shaded regions represent the machined parts of the acrylic. The microscope slide is placed on the top slab, while the magnets are placed in the bottom. The dotted line represents the center of both the microscope slide and magnets, the center of both are aligned.

Pieces of black construction paper were used to improve the contrast of the images and block the reflection of the overhead lights in the ferrofluid. One paper was taped to a tertiary stand and placed behind the acrylic slabs, and another black paper was taped over top of the apparatus. Another stand was set up to hold a flashlight to ensure uniform light over the sample. A simple model of the apparatus set up including the clamps, acrylic slabs, and flashlight, can be seen in Figure 5. The paper is denoted by the thick black line and the shaded region.

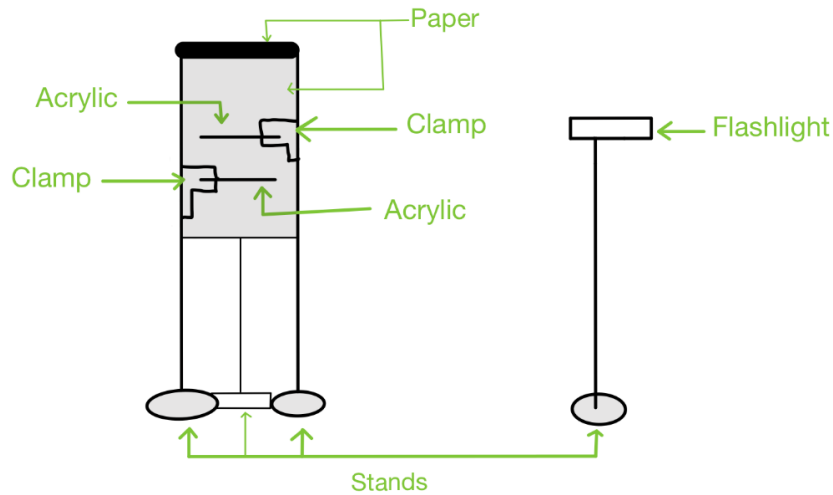


Figure 5: Apparatus set up, the two acrylic slabs used, clamps, flashlight and stands are labeled. The black paper used as a roof and wall are labeled as well, the paper is denoted by the thick black line and the shaded region. Note the paper was placed behind the clamps and acrylic. Note: nothing in the diagram is to scale.

A macro lens was attached to an iPhone to take sufficiently close images of the sample. The phone was placed on a tripod stand, this stand placed at the same distance from the sample throughout the experiment. The phone stand was 6 cm away from the sample. A side view of the set up can be seen in Figure 6. The stands, acrylic slabs (clamps omitted from this diagram for clarity), table, tripod and phone with the macro lens can be seen in the diagram.

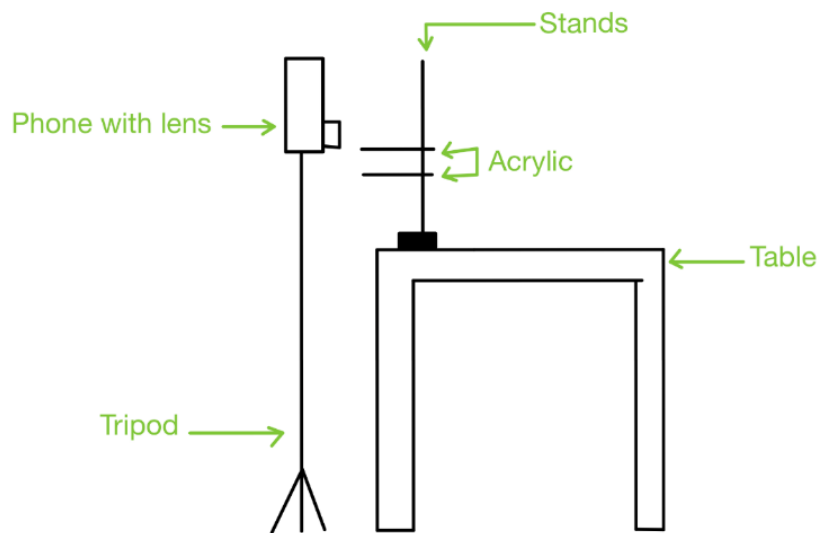


Figure 6: Side view of the set up showing how the camera was placed as well as the placement of the stands on the table. Note: nothing in the diagram is to scale.

A photo of the final setup can be seen in Figure 7 where the clamps and acrylic slabs are labelled. Note that this picture does not include the magnets or microscope slides in their respective slots, but the machined indents can be seen.

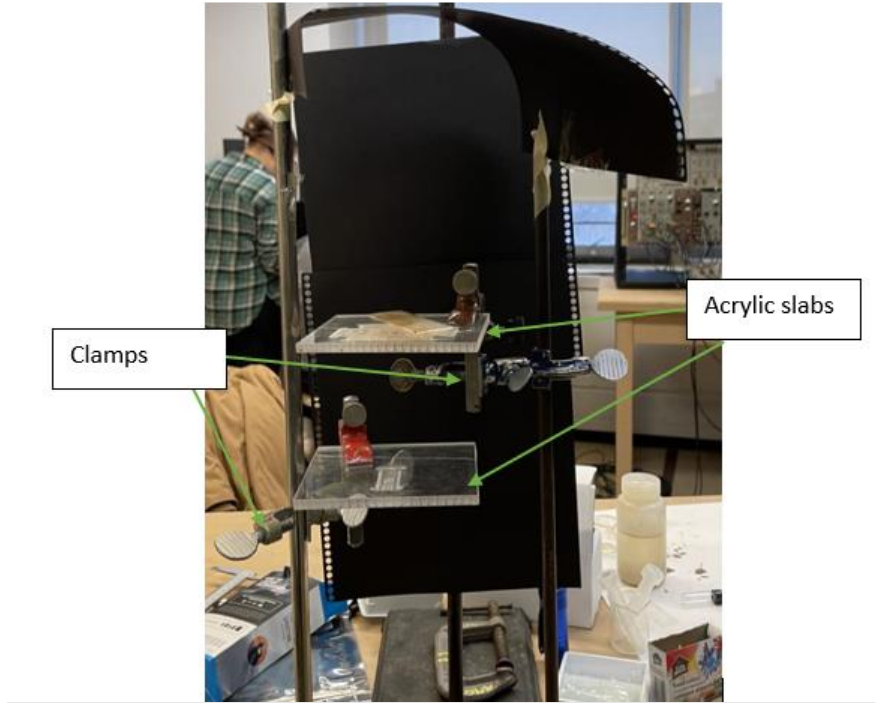


Figure 7: Front view of fully assembled apparatus.

The output voltage of a Linear Hall Effect sensor was attached to an Arduino to measure the magnetic field produced by the neodymium magnets at different distances. The electrical schematic can be seen in Figure 8.

To calibrate the sensor, known values of magnetic intensity were taken using a zero-gauss chamber and a gauss meter alongside varying magnet strength. The linear relation between output voltage and magnetic intensity in mT found is shown in Equation 7.

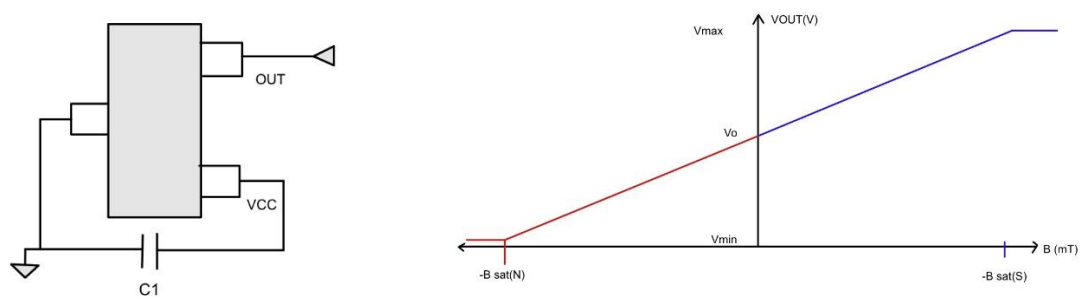


Figure 8: Electrical Schematic and plot of output voltage compared to Magnetic intensity of the Linear Hall Ef

$$B = 0.245 * (V_{out} - 198.5) \text{ mT} \quad (7)$$

The bottom acrylic slab was moved to ten different heights over the experiment, and the magnetic field at each distance was measured. Three different dilutions of ferrofluid were used. One mixture was pure ferrofluid, while the other two samples were diluted using unscented paraffin oil. The

ferrofluid used was the EFH Series ferrofluid [13]. This was chosen because it contains magnetite, is oil soluble and available to be purchased on Amazon [14]. A magnetite ferrofluid was chosen as the referenced papers use magnetite ferrofluid; thus, allowing for a better comparison in results. The paraffin oil was also purchased on Amazon, this oil was chosen since it is a nonpolar oil that has been used in past experiments to dilute ferrofluid [15]. To deposit the ferrofluid onto the microscope slide for imaging a 1 mL syringe was used, each data point is a 0.02 mL sample. A waste bag was set up to dispose of used paper towels and gloves, as well as waste containers where the syringe was cleaned with water and soap.

4. Procedure

4.1 Apparatus Set-up

Once the necessary equipment was gathered, the acrylic was sent to the Stirling Hall machine shop to create the indents where the microscope slides and magnets would be placed to prevent slipping. The Hall Effect sensor has intrinsic properties affecting the range for which it can accurately determine the gradient of the magnetic field, introducing maximum and minimum distances at which the field could be determined. It was found that the closest that the magnet could be to the sensor before malfunctioning (i.e., maximum magnetic strength reading possible) was 1.6cm. The furthest it could be from the magnet was 17.6cm before fluctuations due to natural background noise were higher than the readings. The distance at which the ferrofluid stopped reacting to the magnet used was determined to be around 4 cm. Due to this, data collection was limited to displacements of the magnets below the ferrofluid where the ferrofluid behaviour was still influenced by the magnetic field. The macro lens was tested to observe where the focal length ended. This was done by taking images of a droplet of water with blue food colouring. Once the focal point was determined the tripod was not moved from this position.

The three ferrofluid samples were prepared after determining the three concentration variations to produce an observable difference in the viscosity of the fluid. A pure ferrofluid sample was used, as well as a 1:1 ferrofluid to paraffin oil mixture, and a 1:10 ferrofluid to paraffin oil mixture. The samples were mixed in glass beakers. The 1:1 mixture contained 10 mL of ferrofluid and 10 mL of paraffin oil. The 1:10 mixture contained 3 mL of ferrofluid and 30 mL of paraffin oil. The mixtures were kept in sealed glass containers to limit external particles contaminating the fluid, as well as safety reasons to prevent spills. Every time a mixture was used, the bottle was shaken to ensure the metal particles were evenly distributed in the fluid. Gloves were always worn when handling the ferrofluid to avoid direct contact with the fluid.

4.2 Data Collection

A 1 mL syringe was used to extract 0.020 ± 0.005 mL of each sample. Gloves were worn throughout the experiment and all experimental areas were covered with paper towels to contain any spills. Three people were responsible for data collection, so the person handling the ferrofluid would not cause contamination, and consistent processes were used. The flashlight was turned on at the start of each data collection session. The macro lens was attached to the phone and the phone was set up on the tripod at the same distance from the sample each time. The steps taken were as follows:

1. The acrylic slab holding the magnets was placed at the desired experimental height, starting with the smallest displacement. This height was measured with callipers as the clamps were adjusted to the proper settings, the height was measured three times to ensure the error measurement was minimal. The smallest division in the callipers was taken as the error.
2. A clean microscope slide was placed into the slot in the top acrylic slab.
3. 0.02 mL of the ferrofluid, beginning with the pure ferrofluid concentration, was extracted using the syringe. The sample was placed at the centre of the microscope slide at a consistent location on top of a small indicator on the acrylic.
4. 3-4 images of the sample were taken at full zoom (5x or 132 mm), close care was taken to ensure the images contained the full sample. Often for the glass samples, multiple pictures would have to be taken as the large spread prevented the entire drop from being in focus.
5. The microscope slide was removed and cleaned using a paper towel. If the slide was stained with ferrofluid it was discarded and a fresh one was used. Permanent staining usually occurred after taking 10 samples.
6. Steps 2-5 were repeated using the 1:1 dilution sample, and again with the 1:10 dilution sample.
7. Once all the glass slide measurements were done, steps 2-6 were repeated using the PDMS covered slides. There were two slides being used at a time when conducting the PDMS experiment as the gel had to be applied to the microscope slide.
8. Steps 1-7 were repeated, changing the height of the magnets as specified in Table 3 with the clamps and callipers.

After taking all the images, it was noted that to fit some of the more diluted samples in the image the zoom on the camera had to be adjusted to 2.9x (or 78mm). A set scale is required to complete data analysis on ImageJ, so a picture of the callipers at 5x and 2.9x zoom were taken at the same location where the ferrofluid drops were placed. These were imported to ImageJ during the data analysis portion to ensure the scale of the images was constant. Although special care was taken to ensure the tripod stand did not move, it is likely that it did as the samples were taken over three days and had to be altered slightly to focus. This human error was considered when doing the data analysis as it likely affected the results.

5. Data

Data collection required taking photos of the ferrofluid drops at three different concentrations on both glass and PDMS slides, and altering the distance between the drop and the magnets for 10 height measurements between 21.37 mm and 57.37 mm. The photos were loaded into the ImageJ software for the contact angle, drop height, and drop diameter to be measured. An example of how these measurements were defined is highlighted in Figure 9, with the angle measurement given by the yellow line, and the height measurement given by the blue line. As can be seen in Figure 9 the image of each droplet was not taken at exactly 90 degrees which would have added error to both the height and

contact angle measurement. This experimental error was not accounted for in the calculation of wettability.

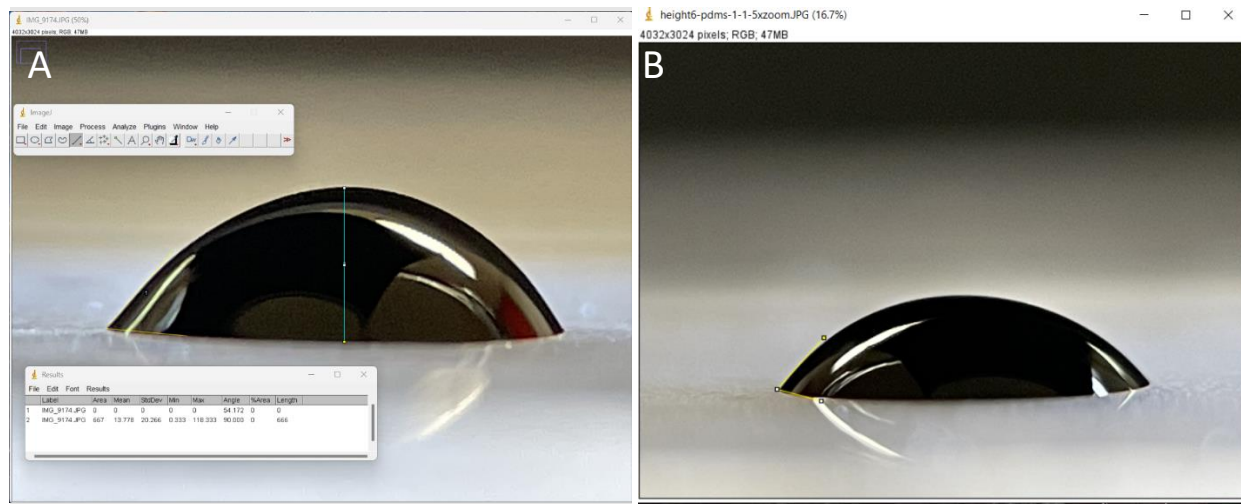


Figure 9: Example of how the height and angle values were measured using the ImageJ software for all trial drops. The height is denoted by the vertical blue line labeled (2) and the angle measurement is denoted by the yellow line on the left corner of the drop labeled (1).

The measurements were performed three times and averaged to minimize human error in the measurement. Furthermore, the magnetic field strength was measured at each of the 10 distances to obtain the characteristics of the magnet that was used in the experiment. Using an Arduino, 125 samples were taken of the magnetic intensity in Tesla at each height and averaged to determine the magnet characteristics. All the data analysis and plotting were completed in Python and Excel, and the associated data tables can be viewed in Table 3 in Appendix 1. The error on each average magnetic intensity reading was determined by using the STDEV.S() integrated function in Excel.

The resulting behaviour of the magnet used in this experiment has been plotted in Figure 10. A power fit for the behaviour was also found since the measured values did not exactly fit the theoretical model.

Experimental and Predicted Magnetic Field of the Neodymium Magnet as a Function of z

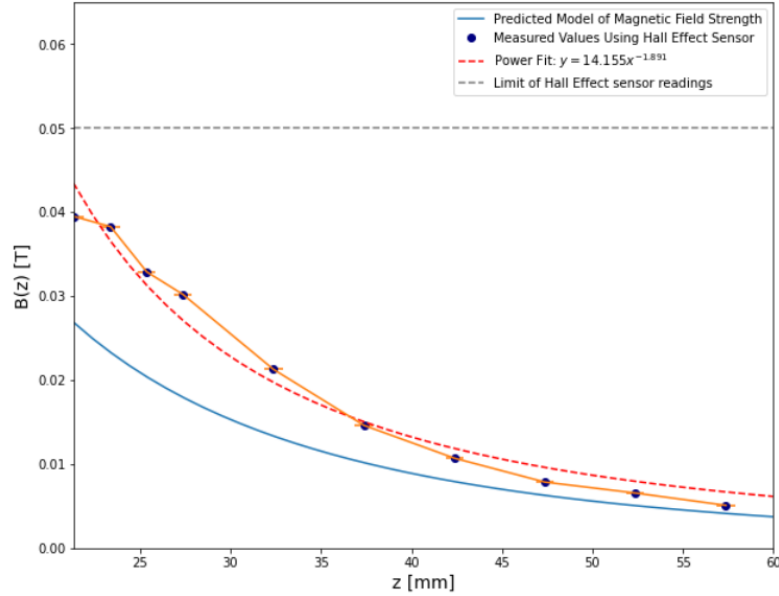


Figure 10: Illustrating the experimentally measured behaviour of the magnet as a function of distance z , along with a fitted power function estimating the function and the original function of a rectangular magnet field strength.

The ferrofluid used in this experiment was provided with a range of magnetite concentrations [13]. As a result, a range of densities, magnetite percentage in solution, and concentrations are summarized in Table 1. Since these datapoints were estimated as an average over a range, this was a source of error to be considered in the analysis for each of the three concentrations.

Table 1: Table with the range of densities, percentages of Magnetite in solution, ferrous concentrations for the three different experimental dilution concentrations used.

Dilution	Percentage of Magnetite of Solution [%]	Density of Solution [g/mL]	Ferrous Concentration of Solution [mol Fe/L]
1:0 Ferrofluid to Paraffin Oil	3-15	1.21-1.42	0.0672-0.394
1:0 Ferrofluid to Paraffin Oil	1.5-7.5	1.01-1.11	0.028-0.154
1:10 Ferrofluid to Paraffin Oil	0.27-1.36	0.84-0.86	0.0042-0.0217

Table 4 through Table 9 in Appendix 1 contain the processed data for both the PDMS and glass surfaces at all concentrations which was collected using ImageJ. Contact angle, drop height, and drop diameter were each measured from three experimental macro photos and averaged to decrease error present in individual photos or measurements. Some images produced, particularly for the 1:1 and 1:10 concentration ratios, were difficult to measure given the ease of spreading on glass material, resulting in extremely flat drops. These datapoints were recorded as zero height and contact angle and were excluded from the data. The remaining datapoints were plotted and can be seen in Figure 11.

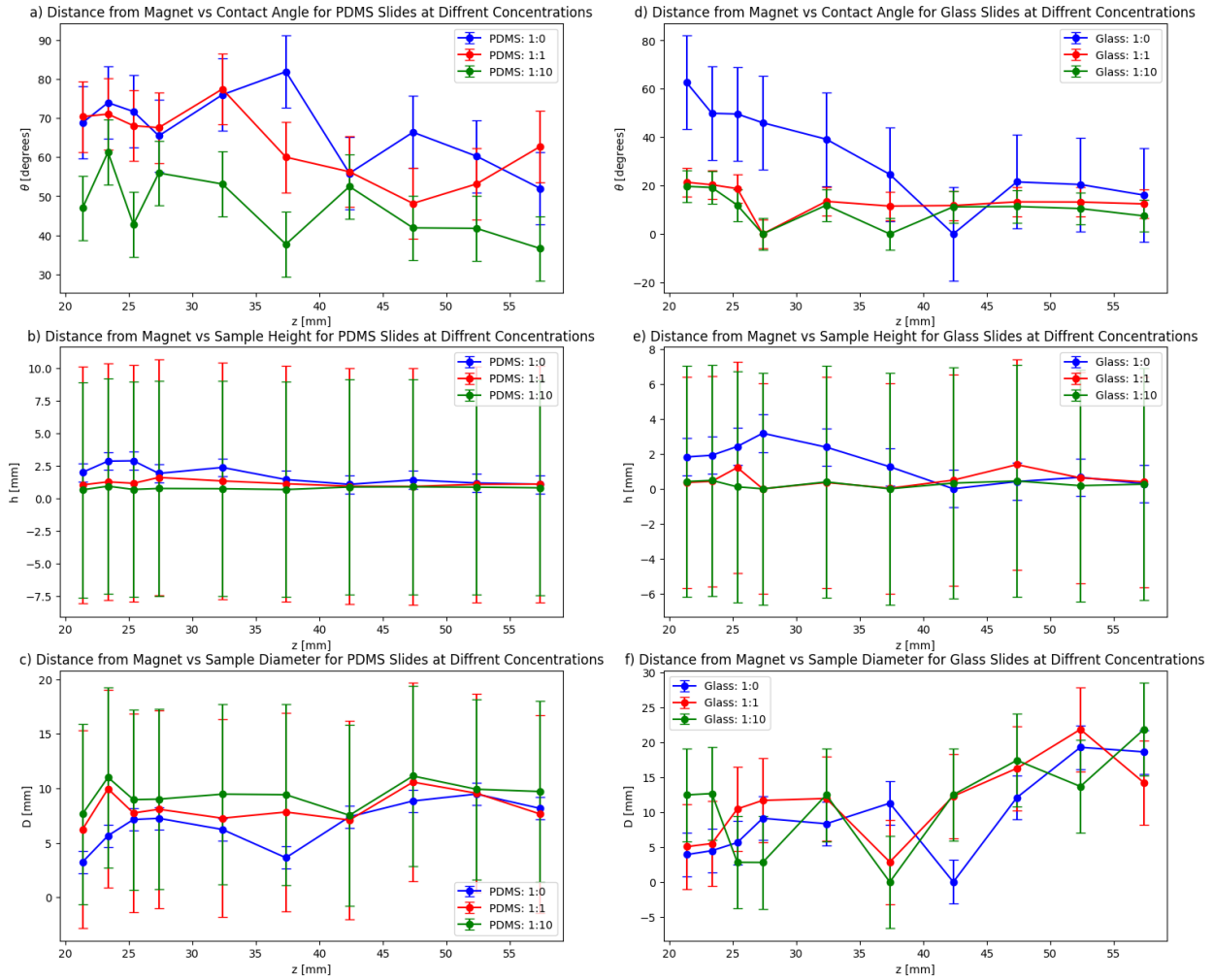


Figure 11: Data collected for the average ferrofluid drop contact angle, height, and diameter for different dilution concentrations, plotted against the distance from the magnets to the ferrofluid. PDMS (left) and glass (right) are compared, note that the glass produced much flatter drops as seen by the height, contact angle, and diameter magnitudes.

6. Analysis

The relationship between contact angle, sample diameter and sample height against magnetic field was explored by plotting and fitting each dataset. The relationship and comparison between PDMS and glass slides as a contact surface is displayed in Figure 12. It is evident that the contact angle decreases as the field strength decreases when using both the PDMS slide and glass slide. This result was unexpected since the contact angle increases with a decreasing magnetic field in past scientific literature [5].

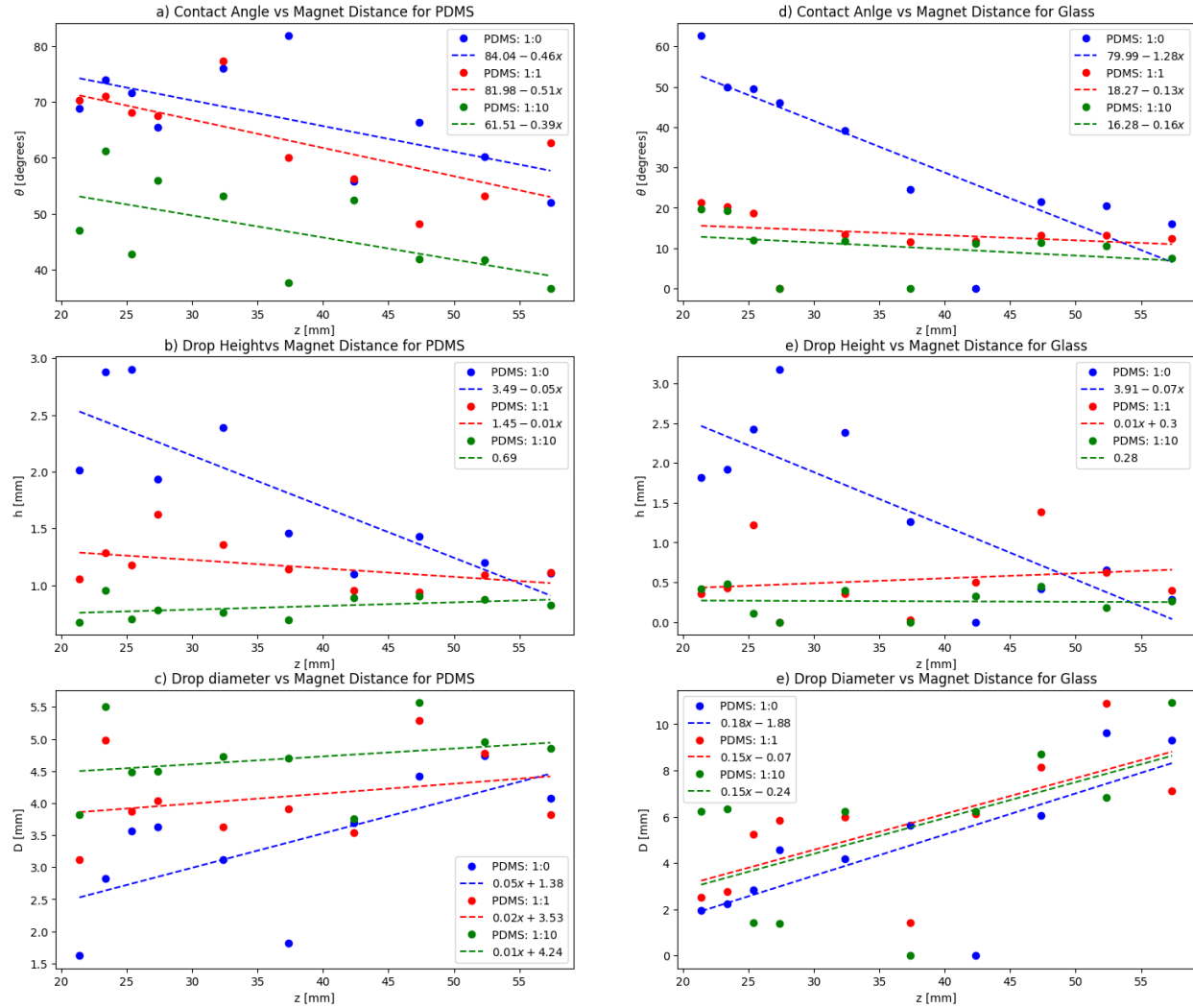


Figure 12: Linear fitted trends of the ferrofluid drop measurements (contact angle, height, and diameter of the drops) on PDMS and glass materials.

Observing the trends for the PDMS surface in Figure 12, all the concentrations appear to decrease at a similar rate for the contact angle as the distance between the sample and the magnets increases. However, on the glass surface, only the 1:1 and 1:10 samples decrease at a similar rate (Figure D and Figure E), while the 1:0 concentration decreases at a greater rate as the distance between the sample and magnets increases (Figure D and Figure E). This can be attributed to the concentration of the ferromagnetic particles present within the different dilution used. The 1:0 sample has the most ferromagnetic particles making it the most susceptible to gradient of the field intensity. For the 1:1 and 1:10 concentrations, the drop height remains relatively stable on both surfaces, while for the 1:0 sample its concentration decreases at a faster rate comparatively as the field strength decreases which is expected. In Figure 12, both the PDMS and glass surfaces follow an increasing trend, where the drop diameter increases as field strength decreases, which was also expected. Although both the glass and PDMS surfaces follow the same general trends, the glass surfaces produce faster and steeper diameter changes than those of the PDMS. This is ascribed to glass having a greater affinity for fluids compared to the PDMS. There is an exception in Figure 12 for concentrations 1:1 and 1:10 where similar trends are

observed for both surfaces. This is the result of the dilution of the ferrofluid with the paraffin oil, resulting in these ferrous concentrations being less susceptible to the gradient of the magnetic field ultimately enabling greater spreading.

There is an exception with the 1:10 concentration but this is likely since the concentration is considerably lower, as displayed in Table 1, that the presence of the magnetic field does not contribute significantly to the contact angle. The drop height decreases as concentration increases which contrasts observations in the experimental results, where a larger concentration of ferrofluid corresponds to an increased drop height. It should be considered that the measurement range was small, and this behaviour is expected to shift at farther distances based on the trend from Figure 12.

Observing the trends for concentration in Figure 12, both material surfaces display a proportional relationship with contact angle and the decreasing concentration of the ferrofluid, which was expected. Figure 12 displays an increase in drop height with higher concentration, which contradicts the expected behaviour an inversely proportional relationship. The last measurement, diameter of the drop, also contradicted expected behaviour as it decreased with increased concentration.

To determine which of the neodymium magnets used in the scientific literature is most comparable to the magnets used in the experimental apparatus, the magnetic curves of the magnets used by the researchers and the power fit of the experimental magnetic curve was plotted as shown in Figure 13. The curves were also constrained to the experimental range of distances to adequately compare the experimental magnetic intensity to *Static Magnetowetting of Ferrofluid Drops* report values [5].

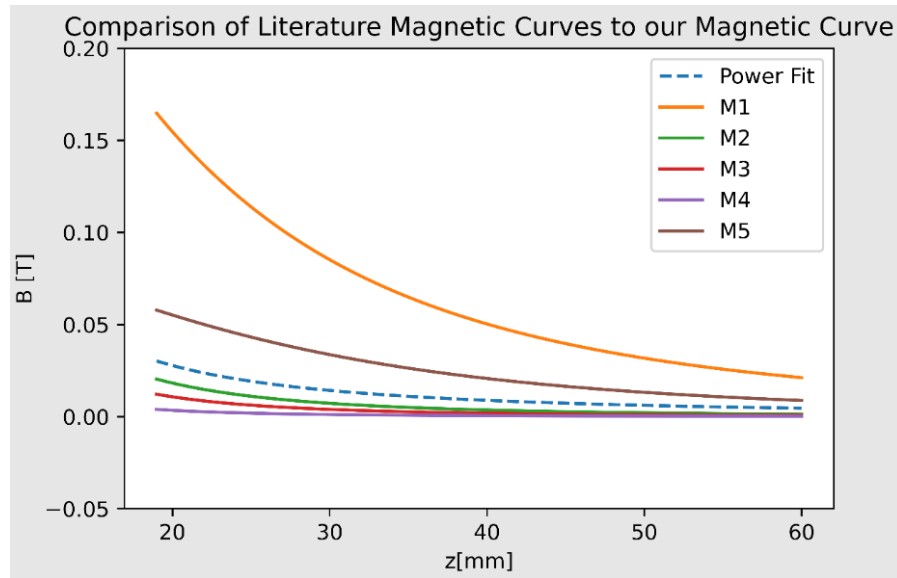


Figure 13: Plot comparing the magnetic curves of the magnets used in the literature to the power fit of the magnet used in this experiment [5].

The results of Figure 13 show that the magnet used in our experiment follows a similar trend to the literature's range of magnets. It can be seen that our magnet is slightly above M2 and below M1. This result is expected since M2 has a maximum magnetic field of $0.50 \pm 0.005 \text{ T}$ and our magnet has a maximum field of $0.309 \pm 0.001 \text{ T}$ which would place its' magnetic curve between M4 and M5 [5]. The properties of all the magnets M1-M5 can be seen in Table 2.

Table 2. Properties of the five neodymium magnets (M1-M5) from the cited literature with Residual Magnetic Field B_r , and Maximum Magnetic Field B_{max} [5].

Magnet	Thickness [mm]	Diameter [mm]	Type	B_r [T]	B_{max} [T]
M1	30.0 ± 0.1	45.0 ± 0.1	N40	1.25	0.48 ± 0.05
M2	20.0 ± 0.1	12.0 ± 0.1	N35	1.17	0.50 ± 0.05
M3	10.0 ± 0.1	10.0 ± 0.1	N45	1.32	0.55 ± 0.05
M4	3.0 ± 0.1	9.0 ± 0.1	N35	1.17	0.33 ± 0.03
M5	5.0 ± 0.1	60.0 ± 0.1	N42	1.29	0.10 ± 0.01

To further check the validity of the experiments results our data for angle, height and diameter was compared to the data from *Static Magnetowetting of Ferrofluid Drops* report [5]. The measurements of the height and diameter are different from our own by an order of magnitude; this is because the researchers used drop sizes of $2.5\mu\text{L}^3$ while due to technical limitations our drop sizes were $20\mu\text{L}^3$. Furthermore, since the data set given is for a sample concentration of 3 mol/L on PDMS then the only valid comparison from our data would be the 1:0 ferrofluid to paraffin oil PDMS concentration as it has a range of 0.0672-0.394 mol/L. The plots from previous experiments at 3 mol/L can be seen in Figure 14.

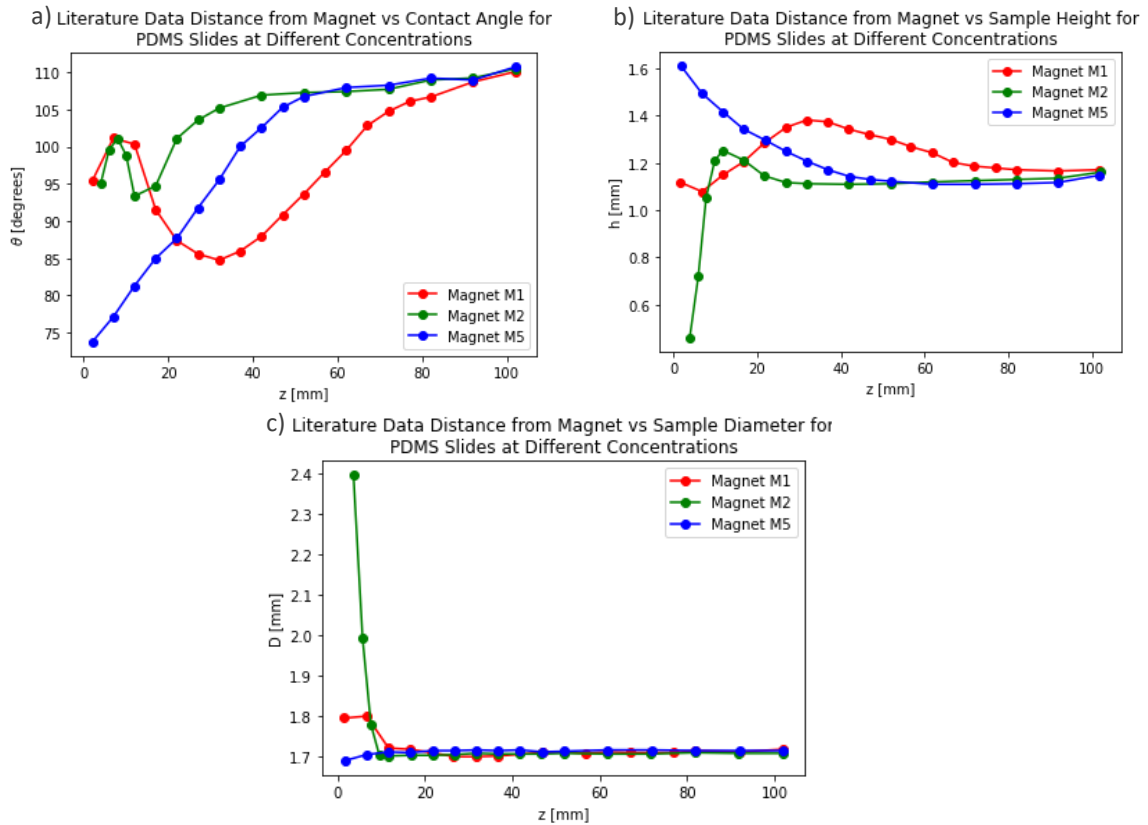


Figure 14: Data plotted in python from the supplied literature data set for the contact angle, the drop height, and the drop diameter against the magnet distance for PDMS slide with a ferrofluid drop of 3mol/L [5].

The plots shown in a) in Figure 14 demonstrate that as magnetic field strength decreases, the contact angle increases, which further confirms that there is an error related to the contact angle measurements. A non-uniform magnetic field causes the contact angle to initially increase with field strength and then it would begin to decrease approximately linearly; thus, it may be the case that for our measurement range we are in the linear decrement portion of the behaviour [5]. This is plausible since the magnet we use is different than the one we are comparing to, meaning that the region of decrement for our magnet could be shifted, causing some discrepancies. The drop heights and sample diameter tend to decrease and converge as the magnetic field strength decreases, as seen in figures b) and c) in Figure 14 respectively.

The data collected during the PDMS trials was plotted against the data from the literature to observe the differences between the datasets and better observe the expected trends. The plot depicting the angles against magnet distance can be seen in graph a in Figure 15. The yellow datapoints are from the literature, while the red, blue and green datapoints are experimental results. The experiment is unable to match the expected angles, this is likely due to errors during data collection and the differences between experimental set ups. No real trend can be observed from the collected data.

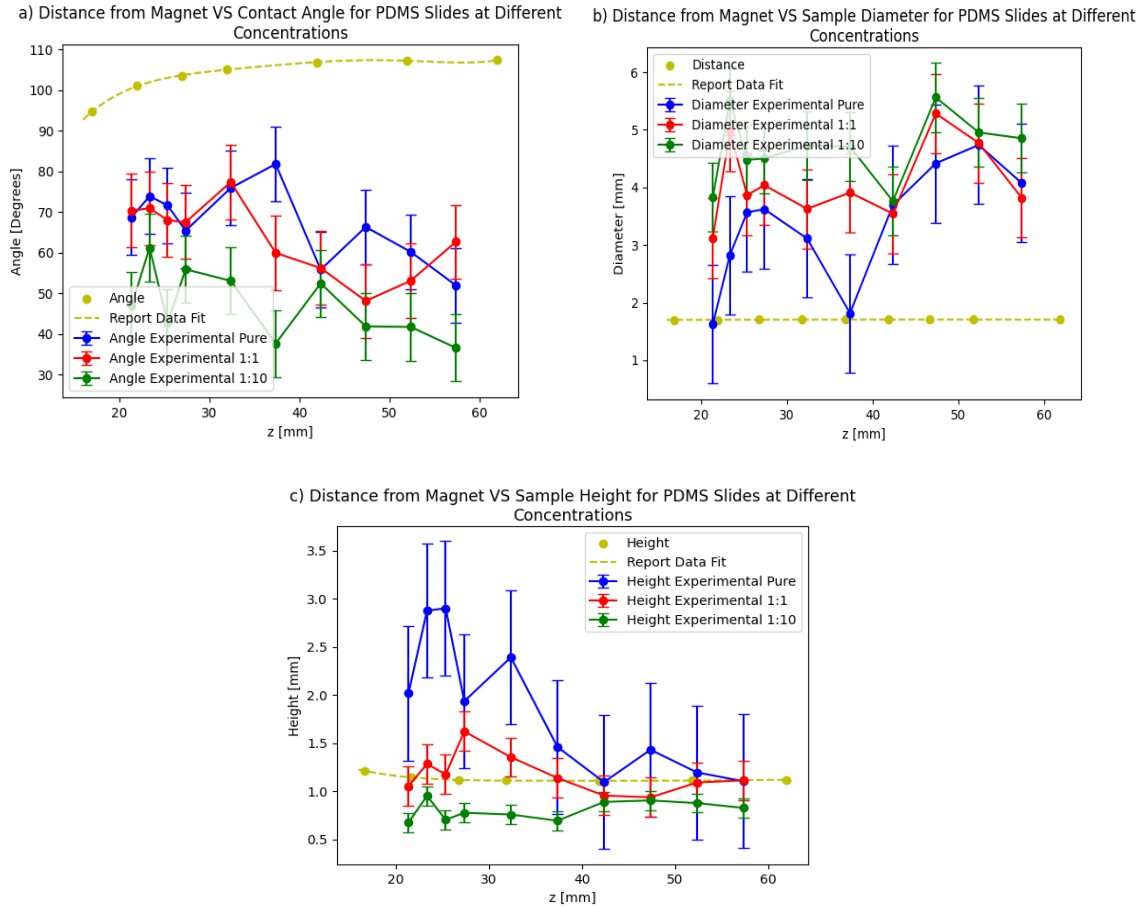


Figure 15: Experimental data (blue, red, green) and literature data (yellow) for a) the angle VS distance from magnet, b) distance from magnet VS sample diameter, c) distance from magnet VS sample height for PDMS slide samples with appropriate error bars [5]

The distance from the magnet and the Diameter of each PDMS sample was examined and compared to the literature in graph b in Figure 15. The dark blue data points are the data from the literature, while the light blue, orange and green are experimental results. The experimental data does not match the data from the literature, however, the data from the pure ferrofluid is sometimes comparable to that of the literature. This could indicate that the pure ferrofluid concentration is comparable to that used in the literature. The converging trend does hold true for the experimental trial. The distance from the magnet and sample height for the PDMS samples was compared to that found in the literature and can be seen in figure c in Figure 15. The experimental data for the 1:1 relation is the most comparable, although as z increases it is evident all the data points begin to follow the decreasing trend found in the literature. Deviation at the start could be from the fact a different ferrofluid concentration was used, and that the magnets used match the literature magnets at higher z values.

Based on Figure 14, drop diameter should increase with an increase in concentration, again contrasting the experimental data from Figure 12. A similar issue with limited range for the trendline from experimental measurements could affect the accuracy of the observed relationship. The comparison between the experimental data, and that from the *Static Magnetowetting of Ferrofluid Drops* report is limited to a trend comparison due to the difference in apparatus and materials used [5]. Both datasets depict the same phenomena, but the measurements differ by an order of magnitude due to experimental set up differences. The quality of the ferrofluid used in this experiment comes with many unknowns, as it was not prepared in a controlled laboratory setting. The ferrous concentration is unknown therefore numerical comparison is not possible. To demonstrate this discrepancy, a χ^2 goodness-of-fit test was conducted. The 1:0 concentration data for the PDMS surface in this experiment from Figure 12 was compared to the trend lines of the sliced data from Figure 15. This yielded the following χ^2 values: $\chi^2_{1:0 \text{ Angle}} = 143.98$, $\chi^2_{1:0 \text{ Height}} = 8.47$, and $\chi^2_{1:0 \text{ Diameter}} = 168.58$. These values of χ^2 are high and show that the experimental data measured does not closely follow the trends found in *Static Magnetowetting of Ferrofluid Drops* [5]. Since the numerical values are not comparable, a qualitative analysis can be conducted by observing trends.

7. Discussion and Future Work

Although the angle versus magnetic intensity for the glass slide and PDMS slides follow a rough trend, the results do not have a strong correlation to the expected results. As mentioned in the analysis the comparisons to the expected results show that the measured data does not accurately describe the wettability of ferrofluid. There are a few factors that differed in our experiment compared to previously conducted experiments. Firstly, the concentration of magnetite in our ferrofluid has a manufacturing range from 3 – 15% while the concentration of the previous report had specifically measured the moles of magnetite for each ferrofluid dilution. Next, there were several aspects of the experimental apparatus that could have caused the inaccurate results. Furthermore, the assumption that the camera (phone with macro lens) was at the exact same distance between the sample for each trial, the alignment of the magnet and the sample by hand, the use of ImageJ to measure the contact angles. To improve the accuracy of the experiment the group could have utilized a procedure that was more precise. For example, an image processing algorithm could have been used with a webcam that was attached to the same stand as the sample. This would mean that the contact angle measurements would have been taken in the exact same fashion for each trial and were not subject to the imprecision of the human eye. As well, if the camera was attached physically to the apparatus the distance from the camera to sample would remain

constant. The tripod used to take the images was not of great quality, it often moved and had to be readjusted for each picture.

Given more time and a larger budget to conduct the experiment a better overall set up could have been produced. 3D printing a set up would have been helpful as the clamps and stands used were not very stable; these often moved and had to be re-adjusted. Creating an apparatus that allowed the camera to be physically attached to the rest of the experiment would have been a better design. Having more precise and higher-quality syringes would have allowed for more precise data collection as the numbering on the syringes would often rub off after being dipped in the samples too many times. Having one syringe designated to type of sample would have also made data collection easier as the syringe had to be cleaned every time a new sample was used. This also caused some of the previous sample to be in the syringe, as well as water and soap. The inclusion of these substances added to the error in the experiment. A better source of light would also help improve the images taken as at times the flashlight did not provide enough light. Having a better method to mix the paraffin oil with the ferrofluid would also help yield better results. If the experiment were to be recreated in the future this is something that should be further researched as it would help provide better, more accurate results. Having access to a tensiometer would also allow us to measure the surface tension of the fluid, which would allow for a better calculation of surface energy. This would allow for a more in-depth analysis to take place and reduce the overall uncertainty.

8. Conclusions

The magnetic intensity of the neodymium magnet matches the expected relation for increases distances to the Hall Effect sensor. However, the contact angle for both PDMS and Glass slides for each dilution of ferrofluid increases as the distance to the magnet decreases which was not expected. As well, the contact angle data has large deviations between trials due to the imprecision of experimental equipment. Furthermore, when comparing the experiments data to the expected results for a 1:0 dilution of ferrofluid, a Chi-Squared of $\chi^2_{1:0 \text{ Angle}} = 143.984$, $\chi^2_{1:0 \text{ Height}} = 8.47$, and $\chi^2_{1:0 \text{ Diameter}} = 168.58$ was found. Since each Chi-Squared value is much greater than 1, there is little to no correlation between this experiments results and the expected values of angle, height, and diameter. Although the Chi-Squared value for the height does show some agreement as it is much closer to 1 than the other values. As well the wettability as expatiated from varying heights and distances for each dilution has a seemingly random distribution, the height and diameter results do follow observed literature trends. The experiment was unable to produce an accurate measure of the relation between magnetic intensity and wettability of the ferrofluids measured.

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12. Supplemental Material

See Supplemental Material for several tables containing the averages of the various data collected for each experimental trial.

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Appendix for: Measuring the Wettability of Different Ferrofluid Concentrations Under the Influence of Magnetic Fields.

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S1. Appendix 1: Tables

Table 3: Shows the values of the average magnetic field strength at a certain distance. Data was recorded with the Hall Effect sensor.

Distance From Magnet to Drop [mm] [$\pm 0.1\text{mm}$]	Average Magnetic Field Strength [mT]	Error of the Average Magnetic Field Strength [$\pm\text{mT}$]
21.37	39.47096	0.21107
23.37	38.27128	0.129633
25.37	32.88496	0.120145
27.37	30.19944	0.141776
32.37	21.2804	0.145543
37.37	14.612	0.120884
42.37	10.70144	0.13481
47.37	7.85648	0.134885
52.37	6.55448	0.133226
57.37	5.08832	0.127695

Table 4: Contains the average drop contact angle, the average drop diameter, and the average drop height for the different distances from the magnet for the concentration of 1:0 PDMS.

PDMS 1:0 Ferrofluid to Paraffin Oil			
Distance From Magnet to Drop [mm] [$\pm 0.1\text{mm}$]	Average Contact Angle [degrees] [$\pm 9.25056\text{degrees}$]	Average Diameter [mm] [$\pm 1.027246\text{mm}$]	Average Height [mm] [$\pm 0.697070\text{mm}$]
21.37	68.80233	1.622334	2.017276
23.37	73.92267	2.82764	2.877794
25.37	71.67133	3.56709	2.90271

27.37	65.45233	3.623454	1.936287
32.37	75.95233	3.11256	2.392342
37.37	81.85567	1.817008	1.461755
42.37	55.87067	3.692924	1.095043
47.37	66.36433	4.41813	1.431902
52.37	60.23700	4.736608	1.195415
57.37	52.02033	4.077928	1.104494

Table 5: Contains the average drop contact angle, the average drop diameter, and the average drop height for the different distances from the magnet for the concentration of 1:1 PDMS.

PDMS 1:1 Ferrofluid to Paraffin Oil			
Distance From Magnet to Drop [mm] [±0.1mm]	Average Contact Angle [degrees] [±9.078134degrees]	Average Diameter [mm] [±0.689542574 mm]	Average Height [mm] [±0.204443889 mm]
21.37	70.35733	3.120322	1.053362
23.37	71.00700	4.976028	1.284666
25.37	68.04367	3.865978	1.174849
27.37	67.53333	4.038576	1.622621
32.37	77.41267	3.628986	1.354943
37.37	59.99100	3.910676	1.139021
42.37	56.25867	3.545126	0.95584
47.37	48.16467	5.282808	0.937187
52.37	53.14100	4.772154	1.091125
57.37	62.69767	3.82212	1.113807

Table 6: Contains the average drop contact angle, the average drop diameter, and the average drop height for the different distances from the magnet for the concentration of 1:10 PDMS.

PDMS 1:10 Ferrofluid to Paraffin Oil			
Distance From Magnet to Drop [mm] [±0.1mm]	Average Contact Angle [degrees] [±8.274795degrees]	Average Diameter [mm] [±0.597093321 mm]	Average Height [mm] [±0.098410387 mm]
21.37	46.99167	3.82388	0.674967
23.37	61.27700	5.500664	0.952612
25.37	42.77833	4.479778	0.703324
27.37	55.94167	4.502302	0.776977
32.37	53.13133	4.730202	0.758542
37.37	37.70133	4.705676	0.693957
42.37	52.48467	3.761624	0.889651
47.37	41.91067	5.562752	0.905179
52.37	41.77933	4.954382	0.876938
57.37	36.63733	4.85324	0.826977

Table 7: Contains the average drop contact angle, the average drop diameter, and the average drop height for the different distances from the magnet for the concentration of 1:0 Glass.

Glass 1:0 Ferrofluid to Paraffin Oil			
Distance From Magnet to Drop [mm] [±0.1mm]	Average Contact Angle [degrees] [±19.39234degrees]	Average Diameter [mm] [±3.122306mm]	Average Height [mm] [±1.071954432 mm]
21.37	62.68200	1.95956	1.821721
23.37	49.91133	2.242354	1.917209
25.37	49.60600	2.821884	2.419489
27.37	45.98333	4.56069	3.177713
32.37	39.18567	4.16908	2.381912
37.37	24.63800	5.63879	1.259882
42.37	0	0	0
47.37	21.52433	6.042172	0.413155
52.37	20.44567	9.636668	0.656578
57.37	16.06667	9.31035	0.287476

Table 8: Contains the average drop contact angle, the average drop diameter, and the average drop height for the different distances from the magnet for the concentration of 1:1 Glass.

Glass 1:1 Ferrofluid to Paraffin Oil			
Distance From Magnet to Drop [mm] [±0.1mm]	Average Contact Angle [degrees] [±6.026578degrees]	Average Diameter [mm] [±2.837512146 mm]	Average Height [mm] [±0.450285666 mm]
21.37	21.39200	2.528998	0.360715
23.37	20.33467	2.753302	0.425742
25.37	18.74300	5.228874	1.215738
27.37	0	5.838846	0
32.37	13.47800	5.975494	0.360057
37.37	11.52467	1.434306	0.029766
42.37	11.75367	6.142942	0.495749
47.37	13.25800	8.132634	1.382285
52.37	13.19500	10.917506	0.625095
57.37	12.42333	7.101718	0.3942

Table 9: Contains the average drop contact angle, the average drop diameter, and the average drop height for the different distances from the magnet for the concentration of 1:10 Glass.

Glass 1:10 Ferrofluid to Paraffin Oil			
Distance From Magnet to Drop [mm] [±0.1mm]	Average Contact Angle [degrees] [±6.62422degrees]	Average Diameter [mm] [±3.455690019 mm]	Average Height [mm] [±3.455690019 mm]
21.37	19.70133	6.2285	0.416379

23.37	19.20133	6.327744	0.479329
25.37	11.91533	1.411852	0.112244
27.37	0	1.40075	0
32.37	11.87000	6.235186	0.392843
37.37	0	0	0
42.37	11.21100	6.247722	0.324879
47.37	11.32933	8.706718	0.445507
52.37	10.49700	6.837236	0.17978
57.37	7.44967	10.919368	0.264877